

SecondSight: A Framework for Cross-Device Augmented Reality Interfaces

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ABSTRACT

This paper describes a modular framework developed to facilitate the design space exploration of cross-device Augmented Reality (AR) interfaces that combine an AR head-mounted display (HMD) with a smartphone. Currently, there is a growing interest in how AR HMDs can be used with smartphones to improve the user's AR experience. In this work, we describe a framework that enables rapid prototyping and evaluation of an interface. Our system enables different modes of interaction, content placement, and simulated AR HMD field of view to assess which combination is best suited to inform future researchers on design recommendations. We provide examples of how the framework could be used to create sample applications, the types of the studies which could be supported, and example results from a simple pilot study.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; **Mixed / augmented reality**.

KEYWORDS

Augmented Reality, smartphone, cross-device interaction, interaction methods, field of view

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1 INTRODUCTION

One of the ongoing challenges with Augmented Reality (AR) using head-mounted displays (HMDs) has been user interaction. Some

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AR HMDs use additional handheld controllers (e.g. the clicker of the Microsoft HoloLens), natural hand gestures (MagicLeap), or touch input on the display (Vuzix Blade), but each of these have limitations.

In recent years companies have begun to develop AR HMDs which can be plugged directly into mobile phones (e.g. the Nreal display). As Zhu et al. [19] point out, smartphones and AR HMDs are complimentary and so a hybrid interface which combines both devices could improve the AR user experience. For example, smartphones provide precise touch screen input compared to the relatively imprecise input in AR HMDs, while AR HMDs support stereo viewing of the AR content, compared to the non-stereo view on a smartphone display. Despite this complimentary nature, there has been relatively little research on how smartphones can be combined with AR HMDs, especially for phone-centric interfaces. This is partly because of the complexity of developing cross-device interfaces and interaction techniques.

In our research, we are interested in exploring how a touch-sensitive smartphone can be used in conjunction with an HMD to enable new types of mobile AR interaction methods and how these are affected by different HMD fields of view (FoV), input techniques, and other factors. For example, commercially available see-through HMDs typically have a diagonal FoV of between 20 and 60 degrees [6], with more recent HMDs also reaching up to 90 degrees (Metavision Meta 2) and 110 degrees (Project NorthStar). As the FoV increases more virtual content can be viewed, which will invariably have an impact on the user behaviour and require different ways of displaying and interacting with the content provided.

To enable exploration of this design space we have created a framework for rapid prototyping of different hybrid interfaces. The main novelty of this research is that our framework enables the exploration of a wider variety of display, interaction and content placement options than previously published research. Overall, design guidelines from user studies based on this framework could provide important insights into future AR HMD smartphone experiences. In the rest of this paper, we provide an outline of the prototype framework, example applications using it, results from a short pilot study and plans for an upcoming evaluation.

2 RELATED WORK

Our research is based on earlier work on AR interaction design and prototyping for hybrid AR interfaces. For the last 50 years, most

AR research has been conducted using HMDs, while over the last 10 years smartphone-based mobile AR has become very popular. However, there has been relatively little research on hybrid AR interfaces which combine handheld and HMD displays in a cross-device interface. For example, a meta-review of HCI papers with hybrid interfaces identified 379 publications [4], but only 3 that were concerned with AR HMDs and mobile phones.

Out of the hybrid AR HMD interfaces which have been presented, most have been HMD-centric, in which AR user interface is designed around mainly viewing content on the AR HMD, while the phone is used for input. For example, the system of Budhiraja et al. [5] views AR content superimposed over the real world in a see-through HMD, and used touch interaction on a mobile phone to interact with the content, with almost no visual cues being shown on the phone. Similarly, the MultiFi system [8] used a see-through AR HMD to view device aligned AR content, and a smartphone or other device to provide input. DesignAR presents a related form of augmented displays which couples a AR HMD with an interactive surface that combines 3D views with 2D information [15].

The research most relevant to ours is the BiShare paper which presents a design space of cross-device interactions between smartphones and AR HMDs [19] but supports both HMD-centric and phone-centric input. Our work is similar in that it supports bidirectional cross-device interaction, but our framework supports a richer range of natural hand gestures, simulation of a wide range of AR HMD fields of view, and has been ported to multiple AR display devices, compared to BiShare which only runs on the Microsoft HoloLens.

Another important topic of research is the effect of AR HMD FoV on the ability to interact with a hybrid interface. Kishishita et al. [10] showed that a larger FoV provides an additional benefit to content interaction, target acquisition and manipulation in AR. Similarly, Blattgerste et al. [3] conducted a study on FoV in AR (with 36, 60, and 90 degrees field of view), simulated in VR, and found that eye-gaze outperforms head-gaze in terms of time-on-task, head movements made and errors in all conditions. Several other studies have shown that by modifying the virtual FoV of presented content in an HMD, performance improved on memorization and learning, as well as target acquisition and content interaction [7, 16]. One of the few studies using a smartphone with an AR HMD with a narrow FoV [2] found that the smartphone screen and the AR content should be able to be viewed simultaneously and that the virtual content should not be placed at the bottom of the smartphone screen. They highlight the currently limited knowledge of joint interaction and visualisation of a smartphone and a AR HMD in a hybrid interface.

This research shows that only recently have researchers begun to explore AR hybrid interfaces that combine a see-through AR HMD with a smartphone. Most of the interfaces that have been developed have been HMD-centric, and there have been few studies exploring interaction methods, and the effect of FoV for cross-device interfaces. The lack of studies shows that there is a need for more research exploring the design space of possible display, interaction techniques and content placement options, and especially tools for rapid prototyping and evaluation of different interface options in hybrid interfaces. In the next section we describe our platform that makes this rapid prototyping possible, and then some example interfaces and an early pilot study with the system.

3 PROTOTYPE SYSTEM

We have developed a prototyping platform for rapidly testing different design ideas for using an AR HMD to "extend" the screen of a smartphone to view several pieces of virtual information simultaneously. The platform has the ability to simulate a range of different HMD FoV to test how AR interface design ideas would be affected by the FoV limitations of current AR headsets. In addition to supporting smartphone touch-screen input, the system also offers two methods to interact with the AR content; free-hand gestures and head-pose. It also allows the placement of virtual AR windows in two different ways: 1) World-fixed, where the user can place the window anywhere in 3D space, and 2) Device-fixed, where the windows are aligned on an invisible 2D plane around the phone. Each of these features has a wide range of parameters that can be specified by the interface designer (e.g. specifying the display FoV), and so the platform provides a high degree of flexibility in terms of the interface combinations that can be explored.

3.1 Implementation

We developed our prototype system on a PC (Intel i7 processor, 64GB RAM, NVIDIA GeForce 1080 TI GPU) running Windows 10 using the Unity game engine (version 2020.1.12). We use the Metavision Meta 2 [9] AR glasses with the Meta Software Development Kit (SDK). The Meta 2 have a resolution of 2560×1440 pixels (1280×1440 per eye) and a wide FoV measuring $82.2^\circ \times 52^\circ$ [11]. We chose the Meta 2 due to it having one of the widest FoV currently available among optical see-through AR HMDs. However the software has also been tested on the Project Northstar HMD [18], and Dreamworld DreamVision HMD [1] and is written in such a way that it would be easy to port to other HMDs. The smartphone used is the Samsung Galaxy S8, although any modern Android phone could be used. The position of the HMD in relation to the phone is tracked via an OptiTrack motion capture system [14] with the HMD and smartphone both having OptiTrack tracking balls attached to them. The HMD and smartphone poses are tracked by ten OptiTrack cameras. Finally, the communication between devices is enabled through the Mirror Networking API [12] for Unity.

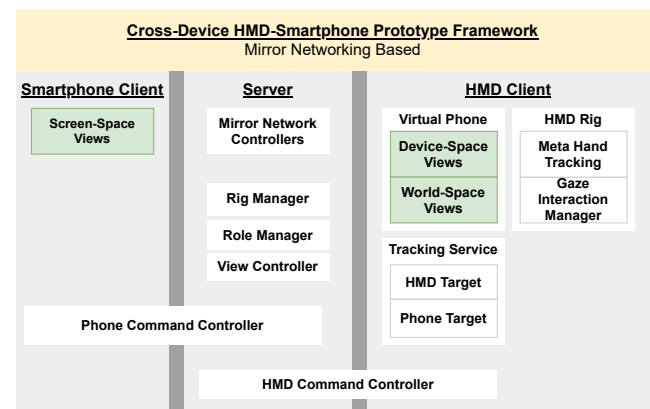


Figure 1: The system components of the rapid prototyping framework.

Figure 1 shows the software framework developed. Software components running on the smartphone and PC with the AR HMD connected, and communicate through a peer-to-peer network that allows for bi-directional interaction between the smartphone and AR HMD. As can be seen, there are three view display components, allowing for information to be shown on the smartphone screen, the AR HMD in device space view (aligned with the smartphone), and the AR HMD in world space view (aligned with the real world). There are also components for hand tracking and head pointing interaction. Finally, the system supports HMD tracking and phone tracking.

A core feature of our application is the dynamic adjustment of the FoV with the intention to emulate different headset experiences. The FoV is simulated via the positioning of several planes around the centre of the user's eye, where a shader renders an unlit black colour on the surface which appear to be transparent on the headset. The position of the planes are based on the distance and angle of the centre point of the user's eyes. The parameters for horizontal and vertical values can be set independently to best emulate existing headset specifications. Figure 2 shows examples of the possible FoV settings.

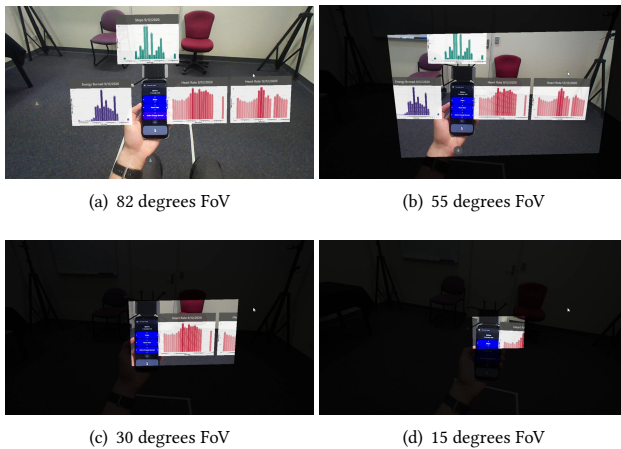


Figure 2: SecondSight emulating different AR HMD FoV.

The framework supports interaction with the content in the AR views in two ways: Gestures and Head-Pose. The user can select the AR content either by directly grabbing it using hand gestures or highlighting it with head pose direction. These two modes were chosen to enable the assessment of the preferences of a "natural" gesture interaction or using the smartphone for input controls. Currently, the user can select, re-position or delete a virtual window around the phone. We use the Meta SDK (version 2.7) for the hand tracking for the gestures. The action to open a window is the same across both interaction modes as it is performed on the phone; the user selects the starting position of the window by swiping the selected content to a desired direction (left, right, up, down). This was adapted from the system of Jie et al. [17], where the user was able to "pull out" a virtual window from a real display. Further, the placement idea was intentionally implemented to identify if participants specific FoVs lead to different behaviours.

Using hand gestures, users can grab a window to re-position it to a desired location or move it to a designated zone below the smartphone, which serves as the "trash bin" to delete the window. For the option using the Head-Pose, a gaze raycast projects a reticle which hovers over the window. The reticle changes colour as the window becomes interactable. Looking at the window, the user can press a designated area on the bottom of the smartphone. Holding the button on the bottom attaches the window to the user's gaze and allows the user to re-position the window. When the button is released, the window will snap in place. Similarly, to delete the window, the user looks at an X icon on the window and uses their phone to complete the action with a single press.

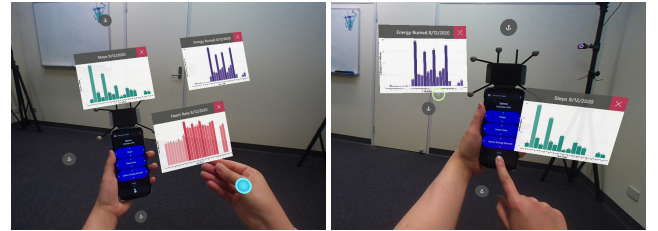


Figure 3: The windows can be selected via hand gestures (left) or by looking at a window and tapping a button on the phone (right).

The work described by Normand et al. [13] presents a scenario of a user being able to watch content co-planar to the phone or have the option to pin the content in space. We are interested in knowing if users prefer to have their windows aligned next to the phone or to have the option to place the windows in space. For this reason, our prototyping system currently supports placing the windows on a 2D plane around the phone (*device-fixed*), or be placed in 3D space at a specified location (*world-fixed*) (see Figure 4).

The device-fixed mode has pre-defined content anchors (referred to as *docks*) on an invisible plane around the phone, which maintains alignment and manages the positioning of the content. As more windows are opened, the system will automatically adjust the windows for best visibility and avoid overlap. In the world-fixed mode, the user can select a position in space around them to place the window. The windows will always rotate towards the user to be visible at all times. The docks allow a variety of types of content to be placed that can easily be replaced depending on the application, such as 3D meshes and 2D images.

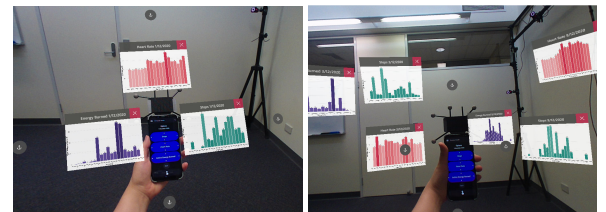


Figure 4: Windows can be arranged on a plane around the phone (left) or in world space (right).

4 EXAMPLE APPLICATIONS

The framework we developed can be used as the basis for a range of different cross-device AR application, each with different interaction techniques. One of the first applications we developed was a phone-centric navigation application. The smartphone is used to view a map view of the user surroundings, while the space around the phone is used to show additional virtual content from the points of interest, viewed on the AR HMD. When the user taps with their finger on a point of interest (e.g., a restaurant), a virtual window appears presented in space showing more information. The user can either use their natural hands, or head pointing to select and position this window in space, either screen aligned with the real phone, or fixed in the real world around the user. When the user selects a point of interest a virtual tag can also appear in the real world, showing where that point is relative to the user. This is a phone-centric interface, so the user is expected to spend most of their attention viewing the real phone screen while using the AR interface for secondary information display.

A second example application is a health visualisation application using an HMD-centric approach. This application is designed to enable the user to view and compare their personal health data over a period of time, including sleep hours, heart rate, steps walked, etc. A simple interface is shown on the smartphone that allows the user to select the day of interest and then select the desired data set from that day that they would like to see. The user can use swipe motions in different directions on the phone screen to create a virtual screen of data aligned with the real phone in that direction. For example, swiping to the left on the heart rate button will show a virtual graph of heart rate data aligned with the left side of the real phone. Once the AR graphs are visible they can be arranged in space around the phone, or placed in the world space using hand gestures or head pointing and phone screen input. For example, a virtual window can be closed by moving the head cursor over the X icon at the corner of the virtual window and tapping on the real phone screen. Figure 5 shows an early-stage implementation of those two example applications.

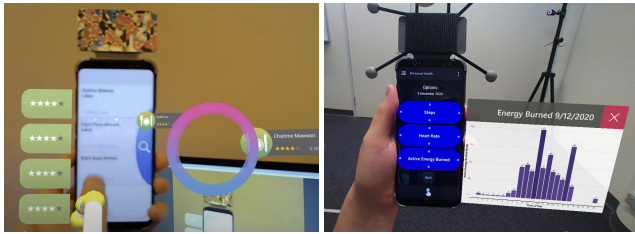


Figure 5: The early prototypes of the AR map interface (left) and an app for simple data visualisation (right).

One of the key reasons for developing these applications is to explore the impact of different interface design choices. As an example, in the next section, we report on a simple pilot study with the health data system to explore the effect of AR HMD FoV on virtual window arrangement.

5 INITIAL PILOT STUDY

We used our system to conduct a short pilot study to assess the effect of AR HMD FoV on the placement of content around the phone in a cross-device interface. We were interested in knowing if we could reproduce similar results as reported by Bang et al. [2], where participants avoided placing virtual content adjacent to the bottom of their phone.

We conduct an initial pilot test with four participants, one female and three male, aged from 25 to 30 years old, and all right-handed. All participants had experience with using AR HMDs, but this was the first time that they had used an AR HMD with a smartphone. After training with the interface, participants were asked to use their phone to open several simplified data visualisations that represented the personal health data of one person. The smartphone app showed a calendar application and each day showed data from three different categories: 1) Heart Rate, 2) Steps, and 3) Active Energy Burned. The data set for the visualisations were taken from one of the investigators personal records, visualised in R and imported as images into Unity.

Participants were asked to open two consecutive days on the calendar and compare each category and report the day with the highest maximum. To place the content, the participant could open the graph for a category by swiping it up, left, right, or to the bottom, with virtual windows of data appearing in the corresponding location attached to the smartphone. Participants completed this task four times using different days. Each time, the simulated AR FoV was set to one of the following horizontal degrees: 15, 30, 55, and 82, in a counterbalanced order. This range of FoVs is typical of a number of commercially available see-through AR displays. The largest FoV matched the FoV of the Meta 2 HMD, 30 degrees is the FoV of the Microsoft HoloLens, 15 degrees is the FoV of the Google Glass, and 55 degrees the FoV of the Nreal display. The same 4:3 aspect ratio was maintained for the other display settings.

After completing the task four times with different FoV settings participants were asked to rate the system on a 5-point Likert scale answering the question *How difficult was it to use the system?* (1 = *Very hard*, 5 = *Very easy*). They were also asked to rank each FoV condition based on preference.

5.1 Results

All four participants had very clear preferences regarding their preferred FoV for this particular task. The widest FoV (82 degrees) ranked highest across all participants, followed by 55 and 30 degrees, and lastly, 15 degrees. This confirmed the answers to the question on difficulty to use, as the widest FoV reached the highest values (see Table 1).

Field of View	How difficult was it to use the system?
15 degrees	1.0 ± 0.0
30 degrees	1.75 ± 0.5
55 degrees	2.75 ± 1.25
82 degrees	3.75 ± 1.25

Table 1: Results for the question on system difficulty.

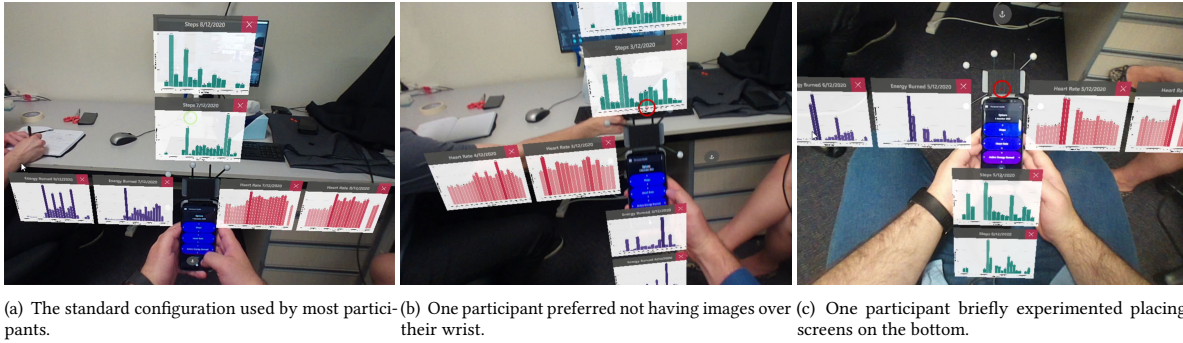


Figure 6: Examples of horizontal and vertical window placements by the user.

All participants arranged their graphs according to a category for an easier comparison, with each category "assigned" a different position (e.g. Heart Rate at the top, Steps to the left). All but one participant arranged their graphs to the left, top, and right. Only one participant placed his graphs at the top, right, and bottom. It is interesting to note that participants' instinctive swiping motion was towards the right during the training session; however, they adapted their approach after being acquainted with the task and from then on always swiped first towards the top of the phone. They commented on avoiding placing virtual content over their wrists.

Interestingly, one of the remaining three participants experimented with the bottom placement for one of the four rounds; however, he switched back to the "standard" placement methods in the next round after expressing dissatisfaction with the bottom placement. See Figure 6 for the participants' configurations. Participants had to move the physical position of the smartphone to be able to inspect the graph in more detail (e.g. closer to the face to "zoom"), which led to two participants saying they preferred having larger images available to avoid these actions.

Currently, the system has limited ways of arranging the content on each side. One participant suggested the idea of a grid system with more options to place content around the phone, as well as providing larger content or the ability to zoom. These ideas will be taken into consideration for the next iteration of our system.

Although a very preliminary pilot study, we were still able to get results that matched the earlier work of [2], as well as getting useful feedback from the subjects on how to improve the interface. This shows the value of our framework for quick prototyping of cross-device interface ideas.

6 FUTURE EVALUATION STUDIES

There are many possible user studies that the platform could be used for, such as exploring the best cross-device interaction techniques, or the impact of using different AR HMDs. We are particularly interested in how the behaviour of the users changes when viewing AR windows depending on FoV, and how participants arrange windows if several are open. Our next step is the execution of a user study employing a mixed-measures experiment design, where the participant will be interacting in two groups. The first group will be able to place windows on a designated plane around the

smartphone device (device-fixed), whereas the other group will be able to place windows freely in space (world-fixed). Each group will undergo 3x2 conditions, using three different FoVs (30, 55, and 82 degrees) and the two previously described interaction methods (gestures and head-pose). As metrics, we will measure task completion time, the position of windows, the direction of preference when opening the window for our objective measures and usability, cognitive load, user preference as subjective measures.

7 CONCLUSION

In this paper, we have described the SecondSight framework that makes it easy to develop cross-device AR systems which combine smartphone display and input with AR HMD viewing. We present the current state of our prototype system and show how it was used to develop several sample applications with different interaction techniques. We also outline results from a small initial pilot FoV experiment to show the types of studies which could be run. The motivation is to make it easy for AR user interface researchers to explore the cross-device design space.

In the future we will use the system to conduct more formal user studies, exploring different display, interaction and content placement factors. We aim to identify user preferences of user interaction and interfaces to build a future set of recommendations that can be applied to different cross-device AR interfaces. Future work will also include additional possibilities of extending the screen, such as combining HMD-aligned, phone screen-aligned and world-aligned content, supporting new interaction methods such as combining voice and gesture input and displaying annotations referring to content on the phone.

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